

Gen-AI training pre-requisites and pre-read

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Overview

This document describes the setup needed to build Gen AI Apps and also gives a primer on Python to get you ready for the Gen AI training.

It is expected to take 2 to 4 hours to go through the below steps depending on your existing familiarity with these topics.

Install and setup Python

1. Install Python

<https://www.python.org/downloads/release/python-3127/>

Use Python 3.12.6 or 3.12.7 only as it is tested with the libraries we will work on
DO NOT INSTALL LATEST PYTHON VERSION

When installing make sure the “Add python to system path” checkbox is selected

2. Test python

Open a command prompt and give the following command:

```
python -version
```

It should show the python version, namely, 3.12.7

If it errors finding the executable, add python.exe location to System environment variable: PATH

You can find the location of python.exe by opening the windows menu, right click on Python app (open file location), copy path

Install and setup VS Code

1. Download and install [VS Code](https://code.visualstudio.com/)

<https://code.visualstudio.com/>

2. Install python extensions in VS Code

1. Open VS Code
2. Install the Python extension:
 - Click the Extensions icon in the sidebar (or press Ctrl+Shift+X)
 - Search for "Python"
 - Install the Microsoft Python extension
 - Other VS Code extensions that are needed only if you want to run the Python notebooks in VS Code (you can always run them in colab)
Jupyter, Jupyter Notebook renderers, ipykernel

Understanding and creating a Virtual Environment for Gen AI projects

Understanding Project Root folder

Project root folder (such as **genaitraining**) or any folder in which you are putting your code, provides the base folder to work against.

When opening VS Code, this is the folder you will open (not any other subfolder)

This is important as your project virtual environment, private configuration (.env) will all reside in your project root folder.

Hence, whenever starting VS Code, **always open the project root folder and not any other folder**

Create a virtual environment for your project

What is Virtual Environment?

Virtual environment facilitates creation of a separate folder where all your python dependencies are saved. This ensures that you do not have to reinstall any library everytime and also that the right versions of your libraries are available to your application code (no conflicts with any other project or setup on your machine)

Following video explains the concept and benefits of virtual environments:

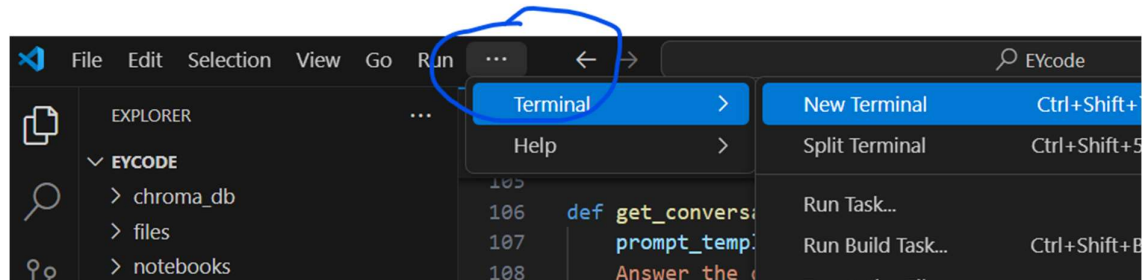
<https://youtu.be/KxvKCSwlUv8?si=6HQHCeJ2wkXnJ0QX>

Follow along the video below or steps outlined below to create a virtual environment

<https://youtu.be/GZbeL5AcTgw?si=cvCzwnbiXPZEWFMA>

1. Create a new folder for your project, called **genaitraining (or code)** in your home directory or in Desktop
2. Open the folder in VS Code (File > Open Folder)

3. Start a terminal



4. In the terminal window, give the following commands to create a new virtual environment:

```
python -m venv venv
```

```
venv\Scripts\activate
```

You will see a popup asking you to use this virtual environment for your project. Click on “Yes”

Install packages needed for Gen AI development

Python package manager **pip** allows you to install all needed packages from a file commonly named as **requirements.txt**

We will create requirements.txt and put all the packages we need for Gen AI development in it. We will then install those packages in our virtual environment.

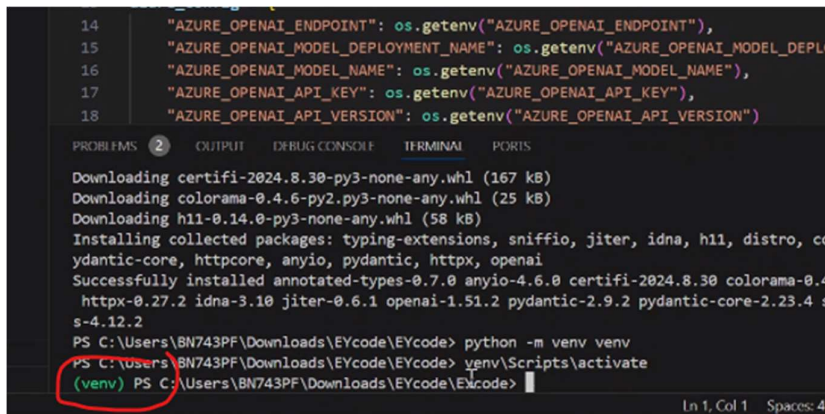
1. In VS Code, with the project root folder (genaitraining or code) selected, click on the +File icon to create a new file, name it “requirements.txt”
2. Put following content in requirements.txt and save the file (ctrl+s):

```
langchain-openai
langchain-community
langchain-text-splitters
chromadb
beautifulsoup4
pypdf
wikipedia
plotly
tqdm
datasets
python-dotenv
openai
streamlit
```

dotted-dict
PyPDF2
matplotlib
seaborn
torch
transformers
tensorflow
tf-keras
jupyter
ipykernel

3. Install packages from requirements.txt:

Make sure your virtual environment is active (you should see the venv name in the command prompt)



The screenshot shows a VS Code terminal window with the following content:

```
14 "AZURE_OPENAI_ENDPOINT": os.getenv("AZURE_OPENAI_ENDPOINT"),
15 "AZURE_OPENAI_MODEL_DEPLOYMENT_NAME": os.getenv("AZURE_OPENAI_MODEL_DEPLO
16 "AZURE_OPENAI_MODEL_NAME": os.getenv("AZURE_OPENAI_MODEL_NAME"),
17 "AZURE_OPENAI_API_KEY": os.getenv("AZURE_OPENAI_API_KEY"),
18 "AZURE_OPENAI_API_VERSION": os.getenv("AZURE_OPENAI_API_VERSION")

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Downloading certifi-2024.8.30-py3-none-any.whl (167 kB)
Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading h11-0.14.0-py3-none-any.whl (58 kB)
Installing collected packages: typing-extensions, sniffio, jiter, idna, h11, distro, co
ydantic-core, httpcore, anyio, pydantic, httpx, openai
Successfully installed annotated-types-0.7.0 anyio-4.6.0 certifi-2024.8.30 colorama-0.4
httpx-0.27.2 idna-3.10 jiter-0.6.1 openai-1.51.2 pydantic-2.9.2 pydantic-core-2.23.4 s
s-4.12.2
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> python -m venv venv
PS C:\Users\BN743PF\Downloads\EYcode\EYcode> venv\Scripts\activate
(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>
```

The terminal window shows the installation of a virtual environment and the activation of the environment. The prompt changes from `PS C:\Users\BN743PF\Downloads\EYcode\EYcode>` to `(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>`, indicating that the virtual environment is active. A red circle highlights the `(venv)` part of the prompt.

Run the following command:

pip install -r requirements.txt

You may see error during the installation mentioning something like below:



The screenshot shows a terminal window with the following content:

```
running bdist_wheel
running build
running build_ext
building 'hnsplib' extension
error: Microsoft Visual C++ 14.0 or greater is required. Get it with "Microsoft C++ Build Tools": https://visualstudio.mic
rosoft.com/visual-cpp-build-tools/
[end of output]

note: This error originates from a subprocess, and is likely not a problem with pip.
```

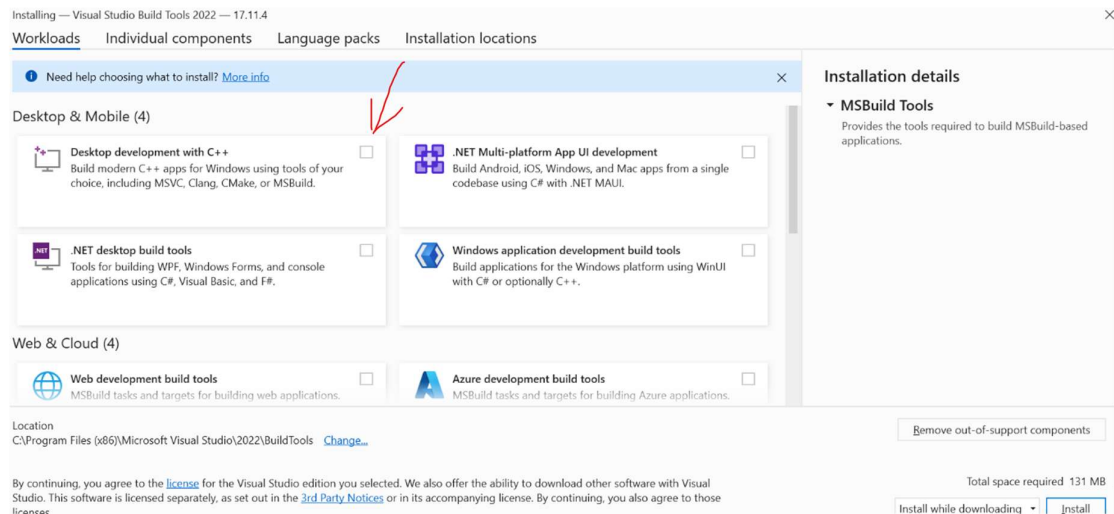
The error message indicates that Microsoft Visual C++ 14.0 or greater is required. The terminal window shows the error message and the URL to download the Microsoft C++ Build Tools.

To fix it, install MS dev tools:

<https://visualstudio.microsoft.com/visual-cpp-build-tools/>

Name	Date modified	Type
▼ Today (5)		
vs_BuildTools.exe	10-10-2024 02:32 PM	Application
tesla_db.zip	10-10-2024 12:24 PM	Compressed (zipped)
tesla-annual-reports.zip	10-10-2024 12:09 PM	Compressed (zipped)
index_tesla_documents.ipynb	10-10-2024 12:00 PM	Jupyter Source File
retrieval_tesla_documents.ipynb	10-10-2024 11:53 AM	Jupyter Source File
▼ Yesterday (6)		

When you open the installer, select the first box below:



After installation, restart VS Code

And open the code folder.

Reactivate the virtual environment in the terminal:

In terminal, execute the following:

```
➤ venv\Scripts\activate
```

Install packages again:

```
pip install -r requirements.txt
```

Troubleshooting

- If terminal says scripts are disabled (PowerShell):

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
```

- If Python is not recognized, verify it's added to PATH
- Make sure to always activate virtual environment before installing packages. The active environment is shown in green as below:

Give command:

`venv\Scripts\activate`

The screenshot shows a VS Code terminal window with a Python script being executed. The script is a simple data processing script that uses `plt.subplots` to create a figure. The output of the script is displayed in the terminal, showing a list of cities and their corresponding population and literacy rates. The terminal prompt is `PS D:\code\EYcode>`. The user has entered the command `venv\Scripts\activate`, which has been executed successfully, resulting in the prompt changing to `(venv) PS D:\code\EYcode>`. The prompt is highlighted in green, indicating that the virtual environment is now active.

- If still facing errors, try following and then run `pip install -r requirements.txt`:
`pip install --upgrade`
`pip setuptools wheel`
`pip cache purge`
`pip install --upgrade --force-reinstall setuptools`

Ater successful installation of packages you should see:

The screenshot shows a VS Code terminal window with the output of a `pip install` command. The output shows the downloading and installation of several packages, including `certifi`, `colorama`, `h11`, `typing-extensions`, `sniffio`, `jiter`, `idna`, `distro`, `annotated-types`, `anyio`, `pydantic`, `httpx`, `openai`, `pydantic-core`, `sniffio`, and `tqdm`. The output also shows the activation of the virtual environment, with the prompt changing from `PS C:\Users\BN743PF\Downloads\EYcode\EYcode>` to `(venv) PS C:\Users\BN743PF\Downloads\EYcode\EYcode>`. The prompt is highlighted in green, indicating that the virtual environment is now active.

Thatsit!!

You are now ready to build your GenAI Python apps with VS code

Python essentials

Download the python_basicapps.zip from:

https://github.com/vijay-agrawal/genaitraining/blob/main/code/00python_basicapps.zip

Unzip it into the **genaitraining** or your project root folder

It contains simple python scripts and a datafile **loanapproval_hist_data.csv**.

Open it in Excel and examine the data.

Loanapproved is the target column (value we want to teach the machine to predict) and other columns are features that influence the target.

Using functions in python

The demo functions_demo.py can be reviewed to understand how function calling, parameter passing is done in Python

Working with data tables using pandas

<https://www.youtube.com/watch?v=tRKeLrwfUgU>

Review the pandas demo code present in pandas_demo.py.

Run it by pressing the play button on top right.

Edit it to play with various dataframe capabilities such as sorting, adding a new derived column etc. Use help of your favourite AI chatbot to generate the code for the same and test it out.

Learn to use .env for API keys and private configuration

Best practice in python based application development is to keep private configuration parameters such as API Keys, passwords etc in a **.env** file

When your python application loads it can read the private key value pairs from the .env file and use them.

This gives flexibility and security and is a common practice to deal with private configuration parameters.

See the video below on how to setup a .env file for your project:

https://www.youtube.com/watch?v=8dlQ_nDE7dQ

Open the dotenv_demo.py in VS Code

Run it by using the “Play” button on top right in your VS Code.

Alternatively, you can open the terminal, activate your virtual environment

➤ Venv\scripts\activate

And issue following command:

➤ python dotenv_demo.py

The script should run and print the environment variables present in your .env file

Working with data files

Data files such as csv files can be easily loaded in Python. They are usually converted to pandas dataframe for further analysis.

Review the file reading demo code present in dataloading.py.

Run it by pressing the play button on top right.

You can check the data file it is loading, present in the data folder.

Using matplotlib for data visualization and analysis

Exploratory Data Analytics (EDA) with python

Run the data_visualization.py to see how matplotlib library helps with data visualization and analysis

A good video on how to do EDA with python:

<https://www.youtube.com/watch?v=xi0vhXFPegw>

Also search on Kaggle for existing notebooks people have made on the same.

<https://www.kaggle.com/code>

Accessing and consuming services using APIs

Review and run the code weather_api_json.py

It uses Weather API service from openweathermap (You can register to get your own free key)

Above should give a good handle on Python essentials for this program

Gen AI Basic app

You will now test a very basic Gen AI Python app, that will use Azure Open AI Keys to access an Open AI LLM hosted on Azure cloud.

Unzip the following into your project root folder:

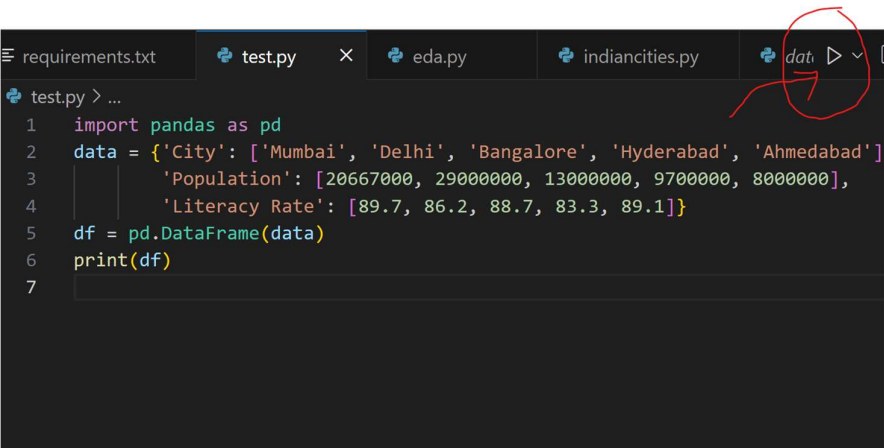
https://github.com/vijay-agrawal/genaitraining/blob/main/code/01azure_openai.zip

Unzip its contents into the genaitraining (or code) folder

Do not unzip into venv (Note that the virtual environment folder (.venv or venv) should not be touched once created)

Make sure you have .env file in your project root folder and it contains the Azure Open AI access keys.

Run the **test_azure_openai.py** by selecting it in the left pane and clicking on the **Play button** or by giving the command in terminal



```
test.py > ...
1 import pandas as pd
2 data = {'City': ['Mumbai', 'Delhi', 'Bangalore', 'Hyderabad', 'Ahmedabad'],
3         'Population': [20667000, 29000000, 13000000, 9700000, 8000000],
4         'Literacy Rate': [89.7, 86.2, 88.7, 83.3, 89.1]}
5 df = pd.DataFrame(data)
6 print(df)
7
```

If unable to run, you may need to select Python Interpreter:

- Press Ctrl+Shift+P (Cmd+Shift+P on macOS)
- Type "Python: Select Interpreter"
- Choose the interpreter from your virtual environment

Also ensure your virtual environment is enabled in the terminal.

In VS Code, open terminal and type the following if not enabled

➤ `venv\scripts\activate`

You should see the output properly showing the response from the LLM

Try to give different prompts such as “Who is the Prime Minister of India” or “Who is the Chief Minister of Maharashtra” or “Tell me about latest results of TCS” etc.

You are now ready to explore the exciting world of AI Apps in the classroom!

Google Colab and Kaggle

Google colab and Kaggle (<https://kaggle.com>) provide a cloud based Python development environment using **Jupyter notebook** concept.

Notebook makes it easy to write and execute python code inline. Below are steps outlined for colab. You can do similar with Kaggle if colab is not workable for you. Kaggle also provides many pre-built notebooks by contributors that are helpful for learning.

Create your first colab notebook

Get started with Google colab by going to:

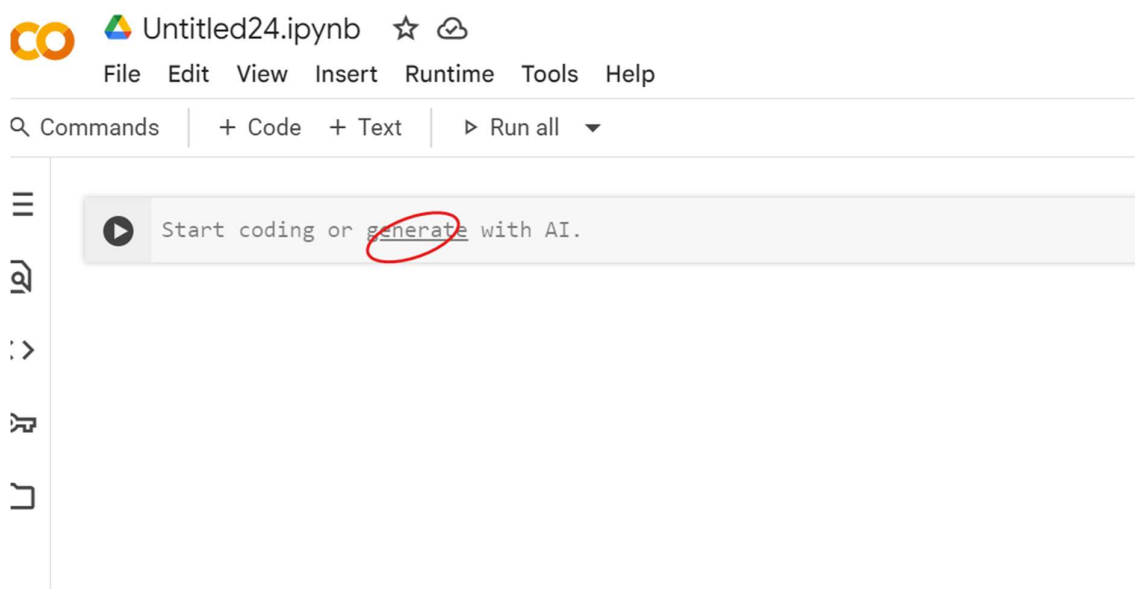
<https://colab.research.google.com/> and registering with your personal gmail account.

Create your first notebook by following the video at:

<https://www.youtube.com/watch?v=RLYoEylHL6A>

Try to put some of the python code from the python demos above into colab and execute them.

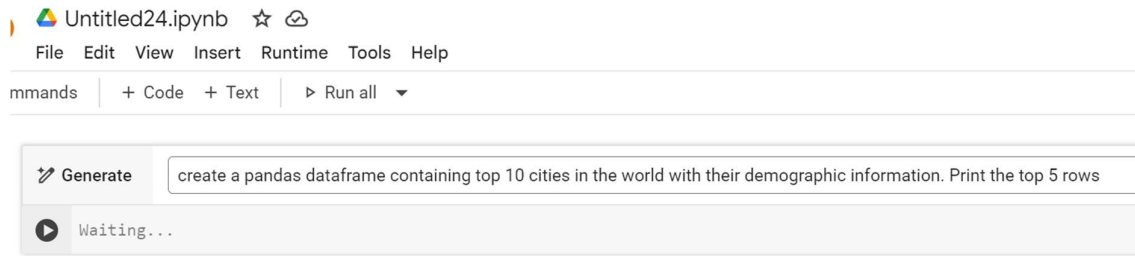
Google colab now also has **code generation** integrated into it!



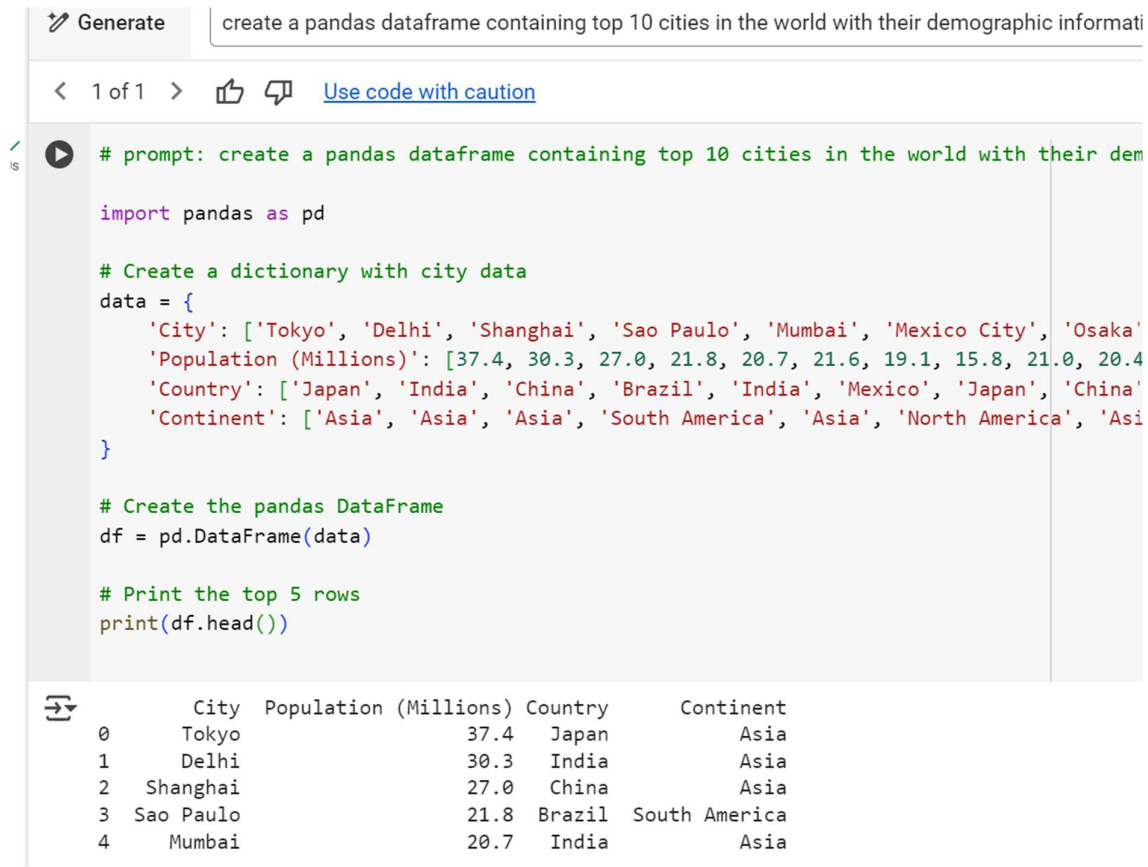
Click on “generate” and you can generate any code you want by just giving the prompt.

For example:

create a pandas dataframe containing top 10 cities in the world with their demographic information. Print the top 5 rows



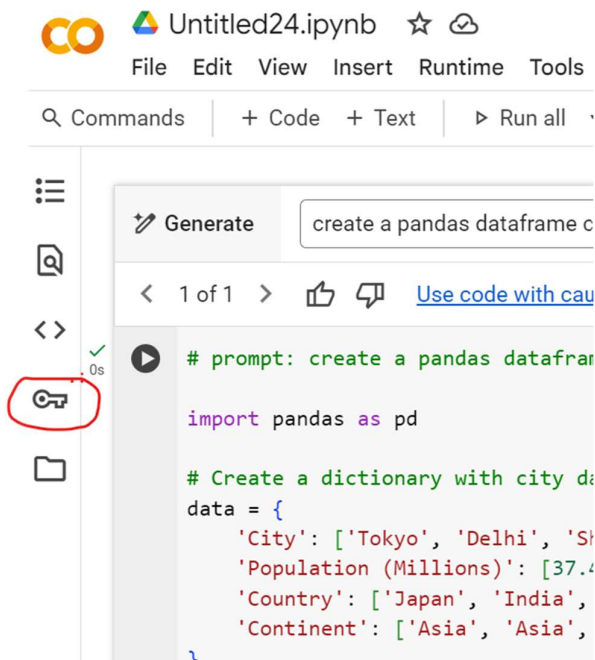
Click on the “Play” button and it will generate the code



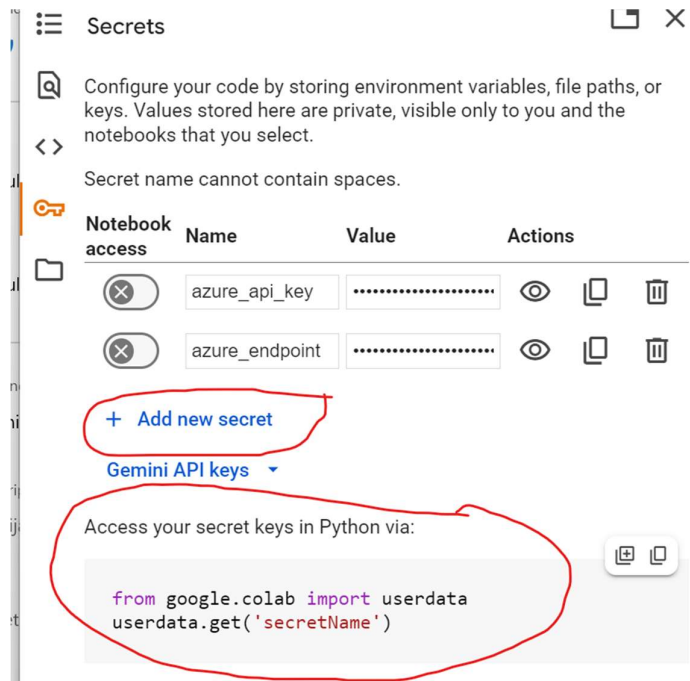
Working with secret keys in Colab

Secret Keys are a way to keep your private configuration information such as API keys secure. (It is similar to the .env concept in your python project in VS Code)

Click on the “Key” icon:



You can add new KEY just like you did in .env file



It shows the code how to use them in your notebook

Copy that code and paste it in your notebook

EXERCISE:

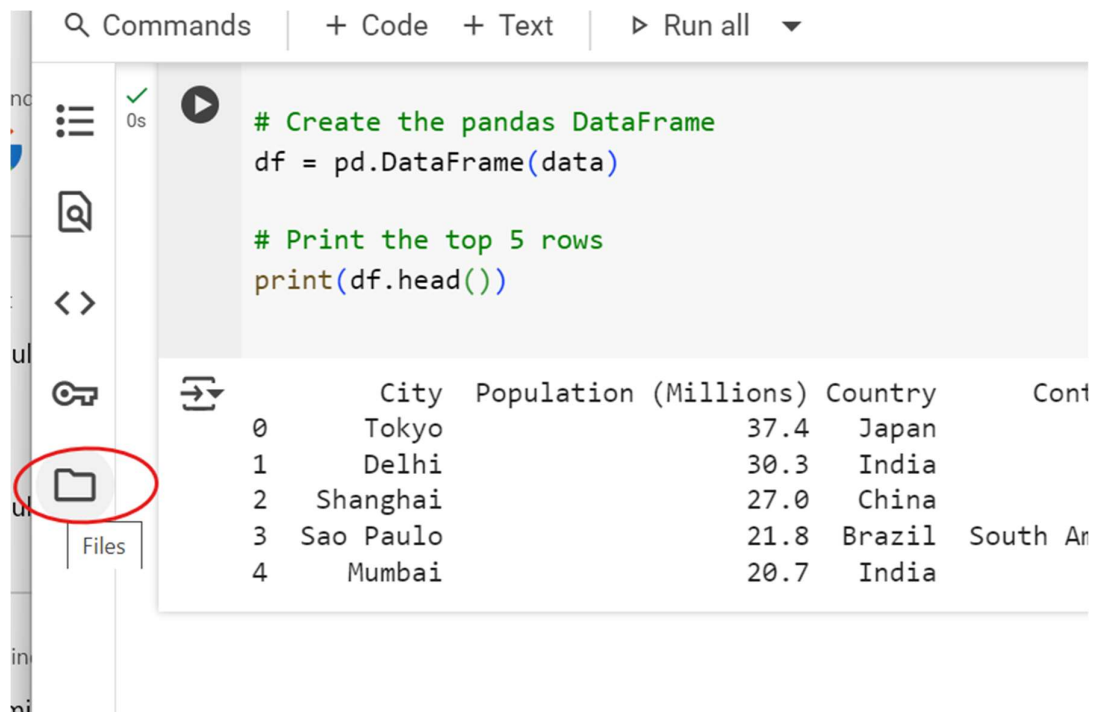
Take the code from the `weather_api_json.py` and make it work in colab.

Hint: remove the `load_dotenv()` and replace with `colabs' userdata.get()` after setting the `OPENWEATHERMAP_API_KEY` secret key in Colab

Working with data files in Colab

Colab offers a filesystem where you can upload your data files and use them in the notebook.

Click on the folder icon on the left



The screenshot shows the Google Colab interface. At the top, there is a toolbar with 'Commands', '+ Code', '+ Text', and 'Run all'. Below this, a code cell is visible with the following Python code:

```
# Create the pandas DataFrame
df = pd.DataFrame(data)

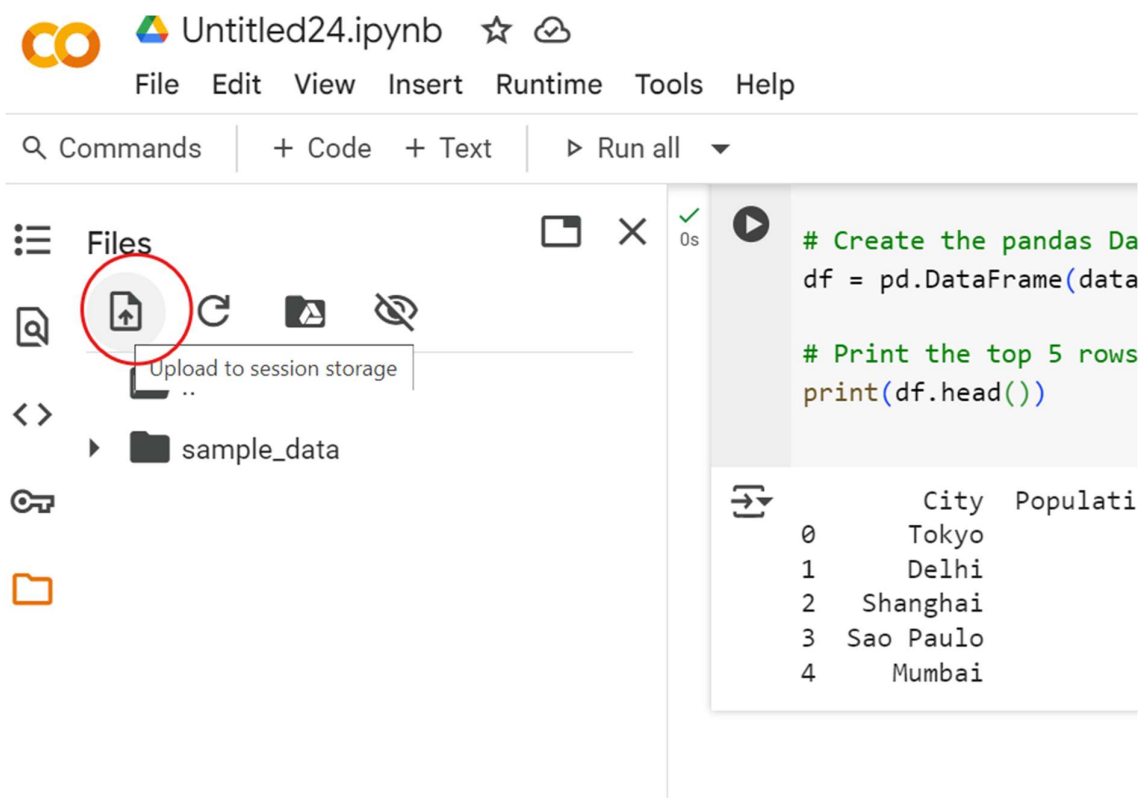
# Print the top 5 rows
print(df.head())
```

Below the code cell, a table of data is displayed:

	City	Population (Millions)	Country	Cont
0	Tokyo	37.4	Japan	
1	Delhi	30.3	India	
2	Shanghai	27.0	China	
3	Sao Paulo	21.8	Brazil	South Ar
4	Mumbai	20.7	India	

On the left side of the interface, there is a sidebar with icons for 'Files', 'Help', and 'Run'. The 'Files' icon, which is a folder symbol, is circled in red.

Click on the "Upload" icon to upload a data file to session storage



Untitled24.ipynb ☆ ☁

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all ▼

Files

Upload to session storage

sample_data

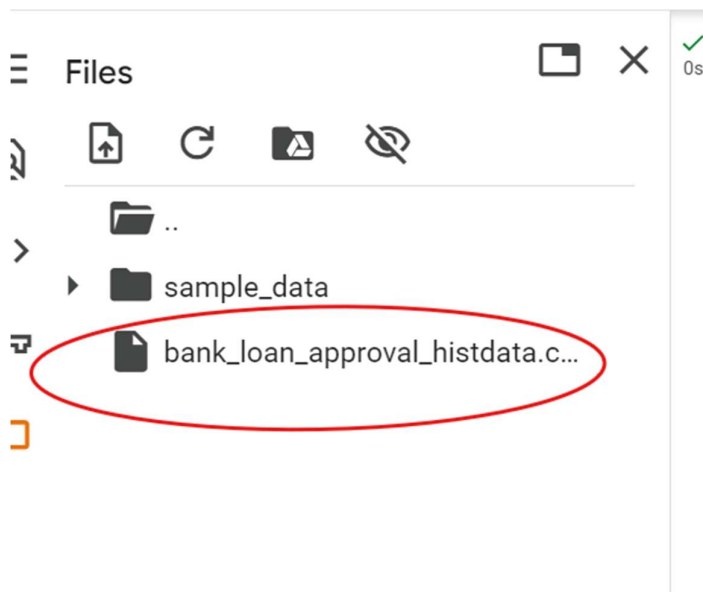
```
# Create the pandas DataFrame
df = pd.DataFrame(data)

# Print the top 5 rows
print(df.head())
```

	City	Population
0	Tokyo	
1	Delhi	
2	Shanghai	
3	Sao Paulo	
4	Mumbai	

Once the file is uploaded it will show up in the left pane.

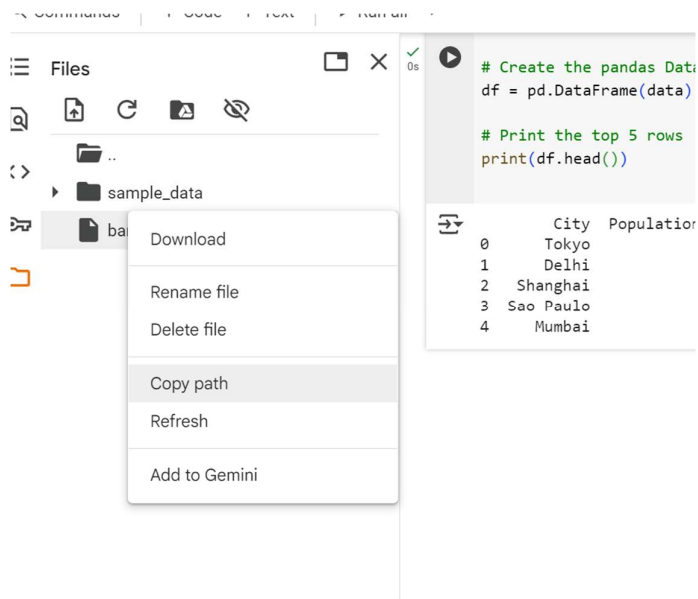
Click on the 3 dots, copy its path



Files

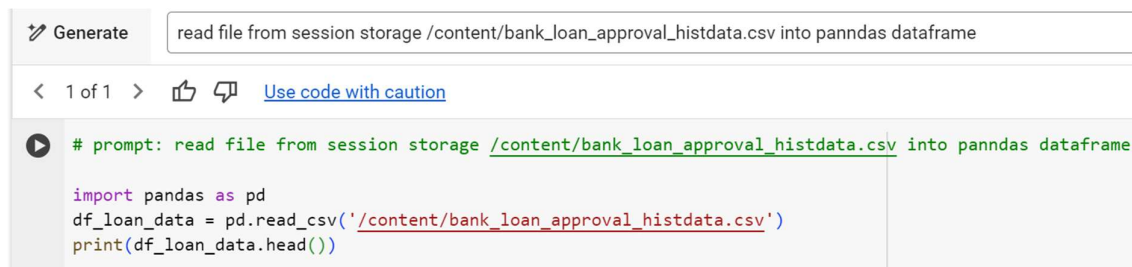
sample_data

bank_loan_approval_histdata.csv



Move mouse toward center of the last cell to see the +Code button. Alternatively you can click on +Code in top level menu

Generate the code as below:



When you run the cell you will see the contents of the file printed

Watch videos below for more information:

Reading csv file:

<https://www.youtube.com/watch?v=VCllZKM7Njk>

How to upload files and use them in colab

<https://youtu.be/58t0PFIWR9Y?si=2oblRNYwyl5sk1Vq>

Git

Git is a version control tool and is needed when interacting your app with Huggingface (HF) Huggingface requires your code to be checked into a HF backed git repository

1. Download Git

1. Visit the official Git website: <https://git-scm.com/download/windows>
2. The download should start automatically for the latest version
3. If it doesn't start automatically, click on the download link for Windows

2. Install Git

1. Run the downloaded installer (git-X.XX.X-64-bit.exe)
2. Adjust your PATH environment:
 - Select "Git from the command line and also from 3rd-party software" (recommended)
3. Complete the installation
4. Note the Path in which git is installed. You may need to add it to the System environment variables if you cannot run git from command prompt

3. Configure Git

1. Open Command Prompt or Git Bash
2. Set your username and email:

```
git config --global user.name "Your Name"
git config --global user.email "your email"
```

Note: When copying pasting the above commands, sometimes spurious characters are getting introduced. Hence manually type these commands if copy paste is not working.

4. Verify Installation

1. Open Command Prompt or Git Bash
2. Type:

```
git --version
```
3. You should see the Git version number if installation was successful

Troubleshooting

- If git commands aren't recognized, try restarting your command prompt or computer
- Ensure git is properly added to your PATH environment variable
- Check firewall settings if you have connectivity issues

For more detailed information and tutorials, visit the official Git documentation:

<https://git-scm.com/doc>

Following videos will help understand git better:

Git overview:

<https://www.youtube.com/watch?v=2ReR1YJrNOM>

Git process and commands:

https://www.youtube.com/watch?v=e9lnsKot_SQ